

| 4. Coordinate systems |                                                                     |                                                                                                                                           |
|-----------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 4.1                   | Cartesian equations for the parabola and rectangular hyperbola.     | Students should be familiar with the equations:<br>$y^2 = 4ax$ or $x = at^2$ , $y = 2at$ and $xy = c^2$ or $x = ct$ , $y = \frac{c}{t}$ . |
| 4.2                   | Idea of parametric equation for parabola and rectangular hyperbola. | The idea of $(at^2, 2at)$ as a general point on the parabola is all that is required.                                                     |
| 4.3                   | The focus-directrix property of the parabola.                       | Concept of focus and directrix and parabola as locus of points equidistant from focus and directrix.                                      |
| 4.4                   | Tangents and normals to these curves.                               | Differentiation of $y = 2a^{\frac{1}{2}}x^{\frac{1}{2}}$ , $y = \frac{c^2}{x}$ .<br><br>Parametric differentiation is not required.       |